

HOMework 4

Continuous-time Fourier Transform (Part 2)

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

- ☐ No AI Used: I did not use any AI tools to complete this homework.
- ☒ AI Assisted: I used AI tools as detailed below:

Where	AI model	How
Question 1.2	Claude	Guiding steps for whether using multiplication or time-shift property first.

Student Acknowledgement: I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

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Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

1. (0.5 points, 0.1 points each) Let:

$$\mathcal{F}\{x(t)\} = X(j\omega) = \text{rect}\left(\frac{\omega-1}{2}\right)$$

Find Fourier transform of

1.1. $x(-2t+4)$

$$\left[x(at+b) = \frac{1}{|a|} e^{j\omega \frac{b}{a}} X\left(j\frac{\omega}{a}\right) \right]$$

$$\begin{aligned} x(-2t+4) &= \frac{1}{|-2|} e^{j\omega \left(\frac{4}{-2}\right)} X\left(j\frac{\omega}{-2}\right) \\ &= \frac{1}{2} e^{-j2\omega} \text{rect}\left(\frac{\frac{\omega}{-2}-1}{2}\right) \end{aligned}$$

$$= \frac{1}{2} e^{-j2\omega} \text{rect}\left(\frac{-\omega-2}{4}\right)$$

Even function
 $[\text{rect}(-x) = \text{rect}(x)];$

$$= \frac{1}{2} e^{-j2\omega} \text{rect}\left(\frac{\omega+2}{4}\right) \quad *$$

* 1.2. $(t-1)x(t-1)$

let $h(t) = tx(t)$

$\therefore h(t-1) = (t-1)x(t-1)$

$$\left[tx(t) \leftrightarrow j \frac{dX(j\omega)}{d\omega} \right]$$

$$\therefore H(j\omega) = j \frac{dX(j\omega)}{d\omega}$$

$$\left[x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega) \right]$$

$$\therefore Y(j\omega) = e^{-j\omega(1)} H(j\omega)$$

$$= e^{-j\omega} j \frac{dX(j\omega)}{d\omega}$$

$$= e^{-j\omega} j [\delta(\omega) - \delta(\omega-2)]$$

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1.3. $t \frac{dx(t)}{dt}$

let $g(t) = \frac{dx(t)}{dt}$

$$\left[\frac{dx(t)}{dt} \leftrightarrow j\omega X(j\omega) \right]$$

$$\therefore G(j\omega) = j\omega X(j\omega)$$

$$\left[tg(t) \leftrightarrow j \frac{d}{d\omega} G(j\omega) \right]$$

$$\therefore t \frac{dx(t)}{dt} = j \frac{d}{d\omega} [j\omega X(j\omega)]$$

$$= j \left[jX(j\omega) + j\omega \left(\frac{dX(j\omega)}{d\omega} \right) \right]$$

$$= -X(j\omega) - \omega \left(\frac{dX(j\omega)}{d\omega} \right)$$

$$= -\text{rect}\left(\frac{\omega-1}{2}\right) - \omega \delta(\omega) + \omega \delta(\omega-2)$$

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1.4. $x(2t-1)\exp[-j2t]$

$$\begin{aligned}
 &\text{let } g(t) = x(2t-1) \\
 &\left[g(at-b) \leftrightarrow \frac{1}{|a|} e^{-j\omega(b/a)} X\left(j\frac{\omega}{a}\right) \right] \\
 &\therefore G(j\omega) = \frac{1}{2} e^{-j\omega/2} X\left(j\frac{\omega}{2}\right) \\
 &\quad \text{a=2, b=1}
 \end{aligned}
 \quad \left| \quad
 \begin{aligned}
 &\left[e^{j\omega_0 t} g(t) \leftrightarrow G(j(\omega-\omega_0)) \right] \\
 &G(j(\omega+2)) = \frac{1}{2} e^{-j\frac{(\omega+2)}{2}} X\left(j\left(\frac{\omega+2}{2}\right)\right) \\
 &= \frac{1}{2} e^{-j\frac{(\omega+2)}{2}} \text{rect}\left(\frac{\omega}{4}\right) \quad \text{※}
 \end{aligned}$$

1.5. $x(t) * x(t-1)$

$$\begin{aligned}
 &\text{let } h(t) = x(t-1) \\
 &\therefore H(j\omega) = e^{-j\omega} X(j\omega) \\
 &\left[x(t) * h(t) = X(j\omega)H(j\omega) \right] \\
 &= X(j\omega) e^{-j\omega} X(j\omega) \\
 &= e^{-j\omega} [X(j\omega)]^2 \\
 &\therefore X(j\omega) \text{ is rect}(\dots) \text{ function, } \text{rect}(\dots)^2 = \text{rect}(\dots) \\
 &\therefore [X(j\omega)]^2 = X(j\omega) \\
 &= e^{-j\omega} \text{rect}\left(\frac{\omega-1}{2}\right) \quad \text{※}
 \end{aligned}$$

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2. (0.5 points, 0.25 points each) For each of the following Fourier transform ($X(j\omega)$), use Fourier transform properties to determine the corresponding time-domain signal ($x(t)$)

2.1. $X(j\omega) = u(\omega) - u(\omega - 2)$

$$\begin{aligned}
 \left[x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{-j\omega t} d\omega \right] & \quad \boxed{t=0} ; \quad x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [u(\omega) - u(\omega-2)] d\omega \\
 & = \frac{1}{2\pi} \int_{-\infty}^{\infty} [u(\omega) - u(\omega-2)] e^{-j\omega t} d\omega \\
 & = \frac{1}{2\pi} \int_0^2 e^{-j\omega t} d\omega \\
 & = \frac{1}{2\pi} \left. \frac{e^{-j\omega t}}{-jt} \right|_0^2 \\
 & = \frac{e^{-j2t} - 1}{2\pi jt} \\
 \text{[Euler's formula]} ; & = \frac{\sin(t)}{\pi t} e^{jt} \quad \text{✖}
 \end{aligned}$$

2.2. $X(j\omega) = \cos(2\omega) \sin(\frac{\omega}{2})$

$$\begin{aligned}
 \text{[Euler's formula]} ; \quad \cos(2\omega) &= \frac{e^{j2\omega} + e^{-j2\omega}}{2} \\
 \sin\left(\frac{\omega}{2}\right) &= \frac{e^{j(\omega/2)} - e^{-j(\omega/2)}}{2j}
 \end{aligned}$$

$$\begin{aligned}
 \therefore X(j\omega) &= \left(\frac{e^{j2\omega} + e^{-j2\omega}}{2} \right) \left(\frac{e^{j(\omega/2)} - e^{-j(\omega/2)}}{2j} \right) \\
 &= \frac{1}{4j} \left[e^{j2\omega} e^{j(\omega/2)} - e^{j2\omega} e^{-j(\omega/2)} + e^{-j2\omega} e^{j(\omega/2)} - e^{-j2\omega} e^{-j(\omega/2)} \right] \\
 &= \frac{1}{4j} \left[e^{j\left(\frac{5}{2}\omega\right)} - e^{j\left(\frac{3}{2}\omega\right)} + e^{-j\left(\frac{3}{2}\omega\right)} - e^{-j\left(\frac{5}{2}\omega\right)} \right]
 \end{aligned}$$

$$\left[\delta(t-t_0) \leftrightarrow e^{-j\omega t_0} \right] ; \quad = \frac{1}{4j} \left[\delta\left(t + \frac{5}{2}\right) - \delta\left(t + \frac{3}{2}\right) + \delta\left(t - \frac{3}{2}\right) - \delta\left(t - \frac{5}{2}\right) \right]$$

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OPTIONAL, FOR PRACTICE ONLY

3. Proof

$$\mathcal{F}\{e^{-|t|}\} = \frac{2}{\omega^2 + 1}$$

4. Using the outcome obtained in Problem 3., Find the Fourier Transform of the given equation.

4.1. $\frac{d}{dt}(e^{-|t|})$

4.2. $\exp(j3t - |2t + 2|)$

4.3. $\frac{1}{2\pi t^2 + 1}$

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5. Proof

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

6. Determine a signal $x(t)$ which has its Fourier transform:

$$X(j\omega) = \delta(\omega) - \delta(\omega - a_0\pi) - \delta(\omega - a_1)$$

And the impulse response in time-domain is $h(t) = u(t) - u(t - 2b_0)$ 6.1. Show how to find $x(t)$ and determine whether it is periodic?6.2. Show how to find $y(t) = x(t) * h(t)$ and determine whether it is periodic?

Verify your results by sketching $x(t)$, $h(t)$, $y(t)$, $X(j\omega)$, $H(j\omega)$, and $Y(j\omega)$. Note that for the complex number, plot only its magnitude.